MILITARY SPECIFICATION

DETONATOR, ELECTRIC, PA537
PARTS FOR
AND
LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, packaging, fabrication, manufacture and loading of one type of electric detonator designated as Detonator, Electric, PA537.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-208 - Ink, Marking Stencil, Opaque (Porous and Non-Porous Surfaces).

MILITARY

MIL-P-116 - Preservation, Packaging, Methods Of
MIL-B-117 - Bag, Sleeve and tubing, Interior Packing
MIL-G-45204 - Gold Plating, Electrodeposited
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Test For
MIL-STD-650 - Explosive Sampling, Inspection and Testing

2.1.2 Other Government documents, drawings, and publications. The following other Government documets, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.7)

U.S. ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
12550926 - Detonator, Electric PA537
9326705 - Microdetonator Clip Assembly
12550826 - Test Circuit
12550757 - Ballistic Environment Test Requirement of M762/M767 Components and Subassemblies.

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

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(Copies of specificatons, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. Components and assemblies shall comply with all requirements specified on Drawing 12550926 all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Moisture content

3.3.1 Lead azide. The moisture content of the lead azide at the loading station at the time of loading the detonators shall not exceed 0.30 percent, when tested as specified in 4.5.2.1.

3.3.2 HMX. The moisture content of the HMX at the time and place of loading the detonators shall not exceed 0.30 percent, when tested as specified in 4.5.2.2.

3.4 Output. The detonator shall not dud, shall fire on the first attempt, and shall make a minimum depression of 0.005 inches in the steel witness plate when tested as specified in 4.5.3.1.

3.5 Waterproofness. The detonators shall be waterproof as evidenced by normal functioning after being immersed in water maintained at 70 degrees Fahrenheit (^{o}F) plus or minus 10°F, at a depth of 2 inches minimum, for not less than 48 hours.

3.6 Environmental

3.6.1 Resistance between electrode and cup, thermal shock and vibration. The detonators shall comply with the resistance requirement of Dwg. 12550926 and with output requirements of para 3.4 after being exposed to thermal shock and vibrations as specified in 4.5.4.1.

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3.6.2 Cold temperature. The detonator assembly shall comply with output requirements of 3.4 when conditioned at +65 ± 5°F for two hours and tested as specified in 4.5.4.2.

3.6.3 Hot temperature. The detonator assembly shall comply with output requirements of 3.4 when conditioned at +165 ± 5°F for two hours and tested as specified in 4.5.4.3.

3.6.4 Shock G. The detonator assembly shall comply with the requirements of 4.5.8 after being subjected to a 30,000 g setback force on specified in 4.5.4.4. The detonator assembly shall be conditioned at 70° ± 10°F for two hours minimum prior to testing.

3.7 Shorting during shipping. The detonator shall be shorted when assembled into the microdetonator clip assembly (Dwg. 9326705) as evidence by a resistance of 0.5 ohm maximum between the detonator electrode and the cup.

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Change of qualified product. After first article inspection, the manufacturer shall notify the purchaser prior to the implementation of any change which may affect performance, quality, reliability or interchangeability. Such notification shall include a thorough description of the proposed change and a suggested test plan designed to demonstrate that the change will not adversely affect performance, quality, reliability or interchangeability and that the changed product will continue to meet the specification requirements. The purchaser will review the proposed change and test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the result shall be submitted to the purchaser for review. Based upon the review the purchaser will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating the change may be manufactured and tested prior to approval, however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser. Changes representative of those which are subject to this requirement are any design change in the manufacturing processing such as:

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a. Package structure and material changes or changes to the geometry.

b. Changes to process flow charts, assembly flow chart and testing flow charts.

c. Conductor materials.

d. Assembly operations move from one line or building to another.

3.10 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. Cleaning methods used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements as specified herein, unless disapproved by the Government. The Govenment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibiltity for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Govermnment to acceptance of defective material.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Cup	9245697	50
Plug Assembly	9245694	150
Bridge Plug and Ignition Assembly	12550787	50
Charged Cup	9257046	50
Detonator Assembly	12550926	380

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078 and Table I herein.

4.3.4 Component qualification. Qualification of the component supplied to this specification is not complete until it has been submitted to and passed additional tests performed by the purchaser. These tests may include 30,000 G acceleration tests, 30,000 RPM spin tests and aging tests to evaluate long term storage characteristics. Coordination of testing and test reporting shall be the responsibility of the purchaser.

4.3.4.1 Re-qualification. Re-qualification of changed component(s) may depend on the result circuit and application analysis conducted as a result of the change(s).

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-D-70826 (AR)

PARAGRAPH / CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Detonator assembly and components		SHEET 1 OF 1		See Below
	Cup (Dwg. 9245697)				
	Examination for defects	50		3.2	4.4.2.1
	Plug Assembly (Dwg. 9245694)				
	Examination for defects	50		3.2	4.4.2.3
	Pull test of electrode	50		3.2	4.5.5
	Insulation resistance between electrode and header	50		3.2	4.5.6
	Bridge Plug and Ignition Assembly (Dwg 12550787)				
	Examination for defects	50		3.2	4.4.2.4
	Detonator Assembly (Dwg. 12550926)				
	Examination for defects	Note1		3.2	4.4.2.6
	Waterproofness	80	80-0-1	3.5	4.5.3.2
	Functioning output, dent test	200	200-0-1	3.4	4.5.3.1
	Environmental Tests				
	a) Resistance between electrode and cup, thermal shock and vibration	25	25-0-1	3.6.1	4.5.4.1
	b) Cold temperature conditioning	25	25-0-1	3.6.2	4.5.4.2
	c) Hot temperature conditioning	25	25-0-1	3.6.3	4.5.4.3
	d) Shock, G test	25	25-0-1	3.6.4	4.5.4.4

NOTES:
1. The entire quantity of detonators required for the tests below shall be subjected to the examination for defects prior to testing.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of electric detonators shall contain:

a. Component parts from one lot interfix number from one manufacturer.

b. HMX from one lot number only.

c. Lead azide of the same type from not more than two lots from the same manufacturer, in accordance with the same specification and specification revision and which have subsequently been subjected to uniformly controlled drying and processing conditions by the manufacturer. If more than one lead azide lot is used the lots shall not differ by more than three serial lot numbers.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70826 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9245697
4.4.2.1	Cup				NEXT HIGHER ASSEMBLY 9257046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None define				
Major					
101	Thickness through bottom		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Concentricity of outside diameter with inside diameter		0.40%	3.2	Gage
105	Dented or distorted		0.40%	3.2	Visual
Minor					
201	Inside diameter, minimum		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70826 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550789
4.4.2.2	Bridgewire				NEXT HIGHER ASSEMBLY 12550788
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Platinum wire diameter		See Note	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1. Five (5) measurements shall be made in the first two (2) feet of wire of each spool, and at least once per shift thereafter.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.3	Plug Assembly		1 1	9245694
				NEXT HIGHER ASSEMBLY: 12550788

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pull test of electrode	50	50-0-1	3.2	4.5.5
102	Insulation resistance test		100%	3.2	4.5.6
103	Outside diameter, max		0.40%	3.2	Gage
104	Bead cracked		0.40%	3.2	Visual
105	Gold plating	Note1	Note 1	3.2	4.5.9
Minor					
201	Length of electrode below surface of header		0.65%	3.2	Gage
202	Thickness of header		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1. Sampling and rejection shall be in accordance with MIL-G-45204.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70826 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550787
4.4.2.4	Bridge Plug and Ignition Assembly				NEXT HIGHER ASSEMBLY 12550926
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spot charge missing		0.40%	3.2	Visual
102	Cracks or voids in spot change		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70826 (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.5	Charged Cup		1 1	9257046
				NEXT HIGHER ASSEMBLY
				12550926
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined			
Major				
101	Height of total charge		0.40%	3.2 Gage
102	Moisture content of lead azide		4.4.3.1	3.3.1 4.5.2.1
103	Moisture content of HMX		4.4.3.1	3.3.2 4.5.2.2
Minor				
201	Evidence of poor workmanship		0.65%	3.10 Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70826 (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.6	Detonator Assembly		1 1	12550926
				NEXT HIGHER ASSEMBLY: Final

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup cracked or split		100%	3.2	Visual
2	Explosive on exterior of cup		100%	3.2	Visual
Major					
101	Length of electrode		0.40%	3.2	Gage
102	Length, crimped end to base		0.40%	3.2	Gage
103	Depth to spherical radius		1.0%	3.2	Gage
104	Large diameter		0.40%	3.2	Gage
105	Spherical radius missing, deformed or scratched		0.40%	3.2	Gage
106	Anvil spherical radius verification		0.40%	3.2	Gage
107	Sealant not present 360°		0.40%	3.2	Visual
108	Decrease of sealant on electrode beyond sealed area		0.40%	3.2	Visual
109	Waterproofness test	50	0-1Note1	3.4	4.5.3.2
110	Functioning output (dent test)	200	1-2Note1	3.5	4.5.3.1
111	Environmental test				
	a)Resistance between electrode and cup		100%	3.2	4.5.4.1
	b)Cold temperature conditioning	25	0-1	3.2	4.5.4.2
	c)Hot temperature conditioning	25	0-1	3.2	4.5.4.3
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1. Any actuator that fails to function on first attempt or fails to produce the required output shall be classed defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70826 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9326705
4.4.2.7	Microdetonator Clip Assembly, (After Assembly of Detonator)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Detonator not shorted		100%	3.7	Current limited ohmmeter (1 milliamp, max.)
Major 101	Detonator electrode protruding above flush of clip assembly		100%	3.7	Visual
102	Detonator cup protruding above flush of clip assembly		100%	3.7	Visual
103	Detonator inverted in clip		100%	3.7	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual
NOTES:					

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70826 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Sealed Bag				See Paragraph 5.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Heat seal test	Note1	Note 1	3.2	4.5.10
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1. Sampling and rejection shall be in accordance with MIL-P-116.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and in 4.4.3.1.

4.4.3.1 Moisture content:

Material	Requirement	Test Paragraph
Lead azide	3.3.1	4.5.2.1
HMX	3.3.2	4.5.2.2

The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one (1) sample of each material from each eight (8) hours of detonator production. Composite samples shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all detonators loaded with the non-conforming material, shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of the MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply.

4.5.2 Moisture content.

4.5.2.1 Lead azide. The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.5.2.2 HMX. The moisture content of HMX shall be determined as specified in 4.5.2.1.

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4.5.3 Functioning output tests.

4.5.3.1 Functioning (steel plate). The detonator assemblies shall be temperature conditioned at 70 ± 10 degrees fahrenheit (°F) for two (2) hours minimum (unless otherwise specified) prior to performing the functioning tests. The tests shall be performed as specified in MIL-STD-331, Test 301.1 except that the witness plate be 1/8 inch minimum and optionally round, 1.22 - .01 dia. Witness plate hardness shall be Rockwell B70 to B95. The base of the detonator shall be placed directly against the center of the witness plate and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to the detonator cup and the other to the electrode. The detonator shall be fired by applying energy from a test circuit made to dwg. 12550826. The depression in the witness block shall be measured using a dial indicator equipped with a probe having a 0.003 ± 0.001 inch spherical radius and in accordance with the requirements of 3.4.

4.5.3.2 Waterproofness output. The detonator assemblies shall be immersed in water at a two (2) to three (3) inch depth for 48 hours minimum. The water temperature shall be maintained at 70 ± 10°F from the immersion period. At the end of the immersion time, the detonator assemblies shall be removed from the water, assembled onto the test fixture and fired by applying energy from a test circuit made of dwg. 12550826. The depression in the witness block shall be measured using a dial indicator equipped with a probe having a 0.003 ± 0.001 inch spherical radius and in accordance with the requirements of 3.4.

The test shall be performed as specified in MIL-STD-331, Test 301.1 except that the witness plate be 1/8 inch minimum and optionally round, 1.22 - .01 dia. Witness plate hardness shall be Rockwell B70 to B95. The base of the detonator shall be placed directly against the center of the witness plate and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to the detonator cup and the other to the electrode.

4.5.4 Environmental tests.

4.5.4.1 Resistance between electrode and cup, thermal shock and vibration. The detonator assemblies shall be tested in the following sequence.

a. Measure resistance between the electrode and cup in accordance with 4.5.8.

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b. Condition for two (2) hours minimum as per 3.6.2.

c. Remove from cold chamber and immediately transfer to the hot chamber.

d. Condition for two (2) hours minimum as per 3.6.3.

e. Remove from hot chamber and immediately transfer to the vibrator.

f. Vibrate for 15 minutes minimum in a sweep covering frequency range of 25 - 500 - 25 cycles per second paralleled to the transverse orthogonal axis with base charge down and a 2 $\pm$ 0.2 g's peak.

g. Measure resistance between electrode and cup in accordance with 4.5.8.

h. The detonators shall be functioned as specified in 4.5.3.1.

4.5.4.2 Cold temperature. The detonator assemblies shall be conditioned to -65 $\pm$ 5°F for two (2) hours minimum. The detonators, maintained at the conditioning temperature, shall be functioned as specified in 4.5.3.1.

4.5.4.3 Hot temperature. The detonator assemblies shall be conditioned to +165 $\pm$ 5°F for two (2) hours minimum. The detonator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.3.1.

4.5.4.4 Shock, G. The detonator assemblies, maintained at the conditioning temperature, shall be subjected to a 30,000g setback force in accordance with the ballistic environment test requirements of M762/M767 components and subassemblies, 12550757. After completion of the test, the detonator assemblies shall be tested as specified in 4.5.8. Test samples shall not be returned to the lot.

4.5.5 Pull test of electrode. The plug assembly shall be assembled to the equipment and tests in accordance with Method 211 of MIL-STD-202. The electrode and header shall be observed for compliance with the requirement. This is a destructive test. Test samples shall not be returned to the lot.

4.5.6 Insulation resistance between the electrode and header. The plug assembly shall be connected to the equipment and the required voltage applied for not less than one (1) nor more than two (2) seconds. The equipment shall be observed to determine compliance with the requirements. Any plug assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

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4.5.7 Resistance between electrode and header. The bridge assembly shall be connected to the equipment and the bridge resistance shall be measured when a current limited to ten (10) milliampere is applied. Any bridge assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.5.8 Resistance between electrode and cup. The detonator assembly shall be connected to the equipment and the resistance shall be measured between the electrode and the cup when a current limited to ten (10) milliampere is applied. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.5.9 Gold plating of pin in plug assembly (See Dwg 9245694 covering detail to Dwg 12550926). Sampling, acceptance criteria and test procedure for this test shall be as specified in MIL-G-45204.

4.5.10 Heat seal test of sealed bag (when required). Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

4.5.11 Tool control for spherical radius. The contractor shall verify at the beginning and end of each detonator lot produced that the anvil used meets the spherical radius requirements of (Dwg. 12550926). Non-compliance with the above requirements shall result in the rejection of the lot.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level B. Detonators shall be packaged in accordance with Code of Federal Regulations Title 49, Para 100-17. In addition, each detonator shall be electrically shorted during shipment and storage by means of a metallic connection from the detonator pin to the cup by assembly into microdetonator clip assembly, Drawing 9326705. When destined for shipment and uncovered storage in excess of 180 days from the loading date shown on the packing box to the date of the assembly into the end item, the following shall be included:

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Each intermediate carton, set-up box or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E of MIL-B-117. The bag shall be closed. All excess air shall be exhausted from within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of MIL-P-116.

5.1.2 Level C. Unless otherwise specified, the detonators shall be packaged in accordance with 5.1.1, except that the barrier bag is not required.

5.2 Packing.

5.2.1 Level B. A suitable number of detonators packaged as specified in 5.1.1, shall be packed in an outer box in accordance with Code of Federal Regulations Title 49: Transportation of Explosives and Other Dangerous Articles, Paragraph 173.66.

5.2.2 Level C. Unless otherwise specified, detonators shall be packed as specified in 5.2.1.

5.3 Marking. Each inner container and intermediate carton set-up box or comparable container shall be printed, stamped or labeled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. (Number) Detonator, Electric, PA 537.
Lot No. Date Packed(Mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to A-A-208.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with Dwg. 8796552. The DOT designation shall be:
Electric Blasting Caps - Handle Carefully (see NOTE).

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e. NOTE: "This number of caps packed in one (1) outer pack shall be shown in the marking".

5.4 Shipping. When detonator from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The detonator assembly covered by the specification is intended for use in munition assemblies.

6.2 Ordering data. See Sections 6.1 and 6.2 of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required to:

Commander
AMCCOM
ATTN: AMSMC-QAR-R(D)
Picatinny Arsenal, New Jersey 07806-5000

This address will be specified on the Contract Data Requirements List, DD Form 1423 in the Contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
AMCCOM
ATTN: AMSMC-QAR-R(D)
Picatinny Arsenal, New Jersey 07806-5000

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.6 Check test for possible deterioration. The following test is designed to prevent the loading of detonators which have deteriorated into a major ammunition item. The test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration. Two hundred (200) detonators shall be selected from each lot for this test.

6.6.1 Functioning. The detonators shall be selected at random from each lot. The detonators shall be tested in the following sequence:

a. Measure the resistance between electrode and cup as specified in 4.5.8. If two (2) or more detonators fail to comply with the resistance requirement specified in Dwg. 12550926, the lot shall be rejected.

b. The detonators that are not classed defective in step "a" shall be functioned as specified in 4.5.3.1 and meet the output requirements of 3.4. If any detonator fails to meet the output requirements, it shall be classed defective. If three or more defectives occur, the lot shall be rejected.

6.7 Subject term (key word) listing.

Moisture content
Resistance
Shock, G
Water proofness

Custodian:
Army-AR

Preparing activity:
Army-AR

(Projec 1375-A407)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-70826 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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